

MINOR PROJECT - 6

PROJECT REPORT

K-MEANS CLUSTERING

Our goal is to segment mall customers into meaningful groups based on spending habits and income.

Access to selected dataset:

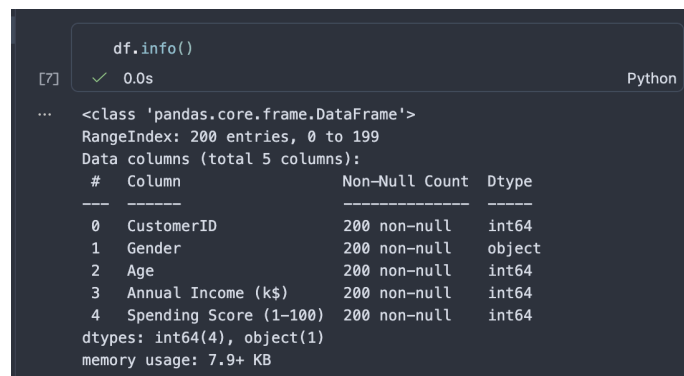
```
import kagglehub

# Download latest version
path =
kagglehub.dataset_download("vjchoudhary7/customer-segmentation-tutorial-in
-python")

print("Path to dataset files:", path)
```

Shape of the dataset : 200 rows, 5 columns

Attributes :



```
df.info()
[7] ✓ 0.0s Python

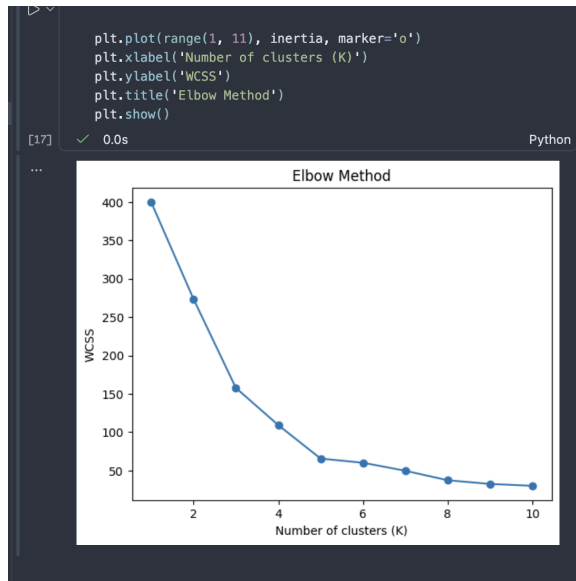
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                   200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Selected features (feature scaling) for K-Means Clustering:

- Annual income in dollars (k\$)
- Spending Score (1 - 100)

The reason:

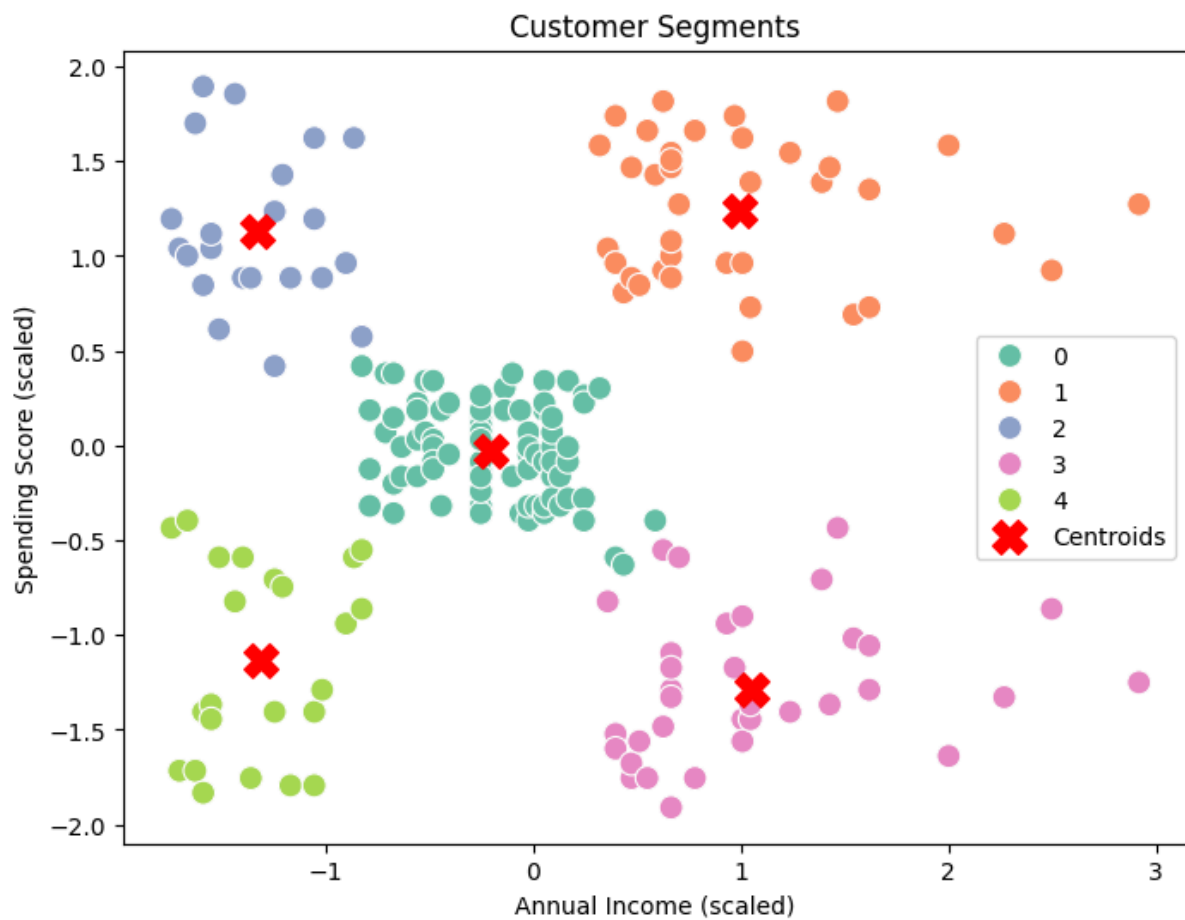
These two features balance financial capacity with actual spending behavior, making them the most meaningful pair for actionable customer segmentation.



Mathematical Justification for selecting k's value as 5:

- Elbow Method:
When plotting WCSS (inertia) vs K, the curve shows a sharp drop until around K=5, after which the reduction in WCSS becomes marginal.
- Silhouette Score
Scores peak around K=5, meaning clusters are well-separated.

Business Insights:



- Cluster 0:
Represents low income but high spending.
- Cluster 1:
Represents high income but low spending.
- Cluster 2:
Represents moderate income and moderate spending.
- Cluster 3:
Represents high-income customers with high spending scores.
- Cluster 4:
Represents low-income customers with low spending scores.